

Microprocessor & Assembly Language

Course No: CSE 3205

1. Components of a Microcomputer

Three main parts of a computer system:

- **Central Processing Unit (CPU)**
- **Memory**
- **I/O Devices**

Working Principle:

1. CPU: Central Processing Unit

Why CPU is the "Brain"

CPU controls the operation of the computer. All instructions are executed in the CPU.

Operation of Control Unit

1. **Fetch:** Fetches instructions from memory that need to be executed.
2. **Decode:** Understands the binary pattern.
3. **Execution:** Executes the decoded instruction.

Operation of ALU:

Microprocessor vs. Microcontroller

Microprocessor:

Incorporates CPU functions on a single IC. (e.g., 8086, 8085). Used in Personal Computers.

Microcontroller:

A compact IC designed to govern a specific operation in an embedded system. (e.g., Washing Machine, Fire detection).

8085 vs 8086

Microprocessor

8085	8086
8-bit processor	16-bit processor
16-bit address bus	20-bit address bus
8-bit data bus	16-bit data bus
Non-pipelined	Pipelined processor
64 KB memory	1 MB memory

Pipelining

The process of fetching the next instruction while the

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Registers of 8086

Data Registers (16-bit / 8-bit):

AX (AH, AL) - Accumulator

BX (BH, BL) - Base Register

CX (CH, CL) - Count
Register

DX (DH, DL) - Data Register

Segment Registers (16-bit):

CS - Code Segment

DS - Data Segment

SS - Stack Segment

ES - Extra Segment

Pointers & Index (16-bit).

Basic Instructions (8086)

Data Transfer:

- MOV dest, src (Transfer data)
- PUSH / POP (Stack operations)
- XCHG (Exchange data)

Arithmetic:

- ADD / SUB (Add / Subtract)
- INC / DEC (Increment / Decrement by 1)
- MUL / DIV (Multiply /

Flag Register (8086)

Processor state is implemented as 9 bits called flags (Status & Control flags).

Status Flags:

- CF (Carry Flag): 1 if carry out from MSB.
- PF (Parity Flag): 1 if even parity.
- AF (Auxiliary Flag): Carry out from bit 3.
- ZF (Zero Flag): 1 if zero result.
- SF (Sign Flag): 1 if MSB is